

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("quotes_dataset.csv")

print("First 5 Rows:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns)

print("\nData Types:")
print(df.dtypes)

print("\nDataset Information:")
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())

plt.figure(figsize=(8,5))
sns.heatmap(df.isnull(), cbar=False)
plt.title("Missing Values Heatmap")
plt.show()
```

```
print("\nStatistical Summary:")
print(df.describe(include='all'))

print("\nSample Questions:")
print("1. Which category appears most frequently?")
print("2. Are there missing values?")
print("3. What patterns exist in the dataset?")
print("4. Are there any unusual observations?")

print("\nDuplicate Rows:", df.duplicated().sum())

num_cols = df.select_dtypes(include=np.number).columns

for col in num_cols:
    plt.figure(figsize=(6,4))
    sns.histplot(df[col], kde=True)
    plt.title(f"Distribution of {col}")
    plt.show()

if len(num_cols) > 1:
    plt.figure(figsize=(8,6))
    sns.heatmap(df[num_cols].corr(),
                annot=True,
                cmap="coolwarm")
    plt.title("Correlation Matrix")
    plt.show()
```

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for col in num_cols:
    plt.figure(figsize=(6,4))
    sns.boxplot(x=df[col])
    plt.title(f"Outlier Detection - {col}")
    plt.show()

from scipy.stats import ttest_ind

if len(num_cols) > 0:
    col = num_cols[0]

    median_value = df[col].median()

    group1 = df[df[col] <= median_value][col]
    group2 = df[df[col] > median_value][col]

    t_stat, p_value = ttest_ind(group1, group2)

    print("\nHypothesis Testing")
    print("T-statistic:", t_stat)
    print("P-value:", p_value)

    if p_value < 0.05:
        print("Significant Difference Found")
    else:
        print("No Significant Difference Found")

print("\nData Quality Report")
print("-----")
print("Missing Values:")

```

```
print(df.isnull().sum())
```

```
print("\nDuplicate Records:")  
print(df.duplicated().sum())
```

```
print("\nData Types:")  
print(df.dtypes)
```

```
print("\nEDA Completed Successfully!")
```

First 5 Rows:

	Quote	Author
0	"The world as we have created it is a process ...	Albert Einstein
1	"It is our choices, Harry, that show what we t...	J.K. Rowling
2	"There are only two ways to live your life. On...	Albert Einstein
3	"The person, be it gentleman or lady, who has ...	Jane Austen
4	"Imperfection is beauty, madness is genius and...	Marilyn Monroe

Dataset Shape:

(10, 2)

Column Names:

Index(['Quote', 'Author'], dtype='object')

Data Types:

Quote object

Author object

dtype: object

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 2 columns):

#	Column	Non-Null Count	Dtype
0	Quote	10 non-null	object
1	Author	10 non-null	object

dtypes: object(2)

memory usage: 292.0+ bytes

None

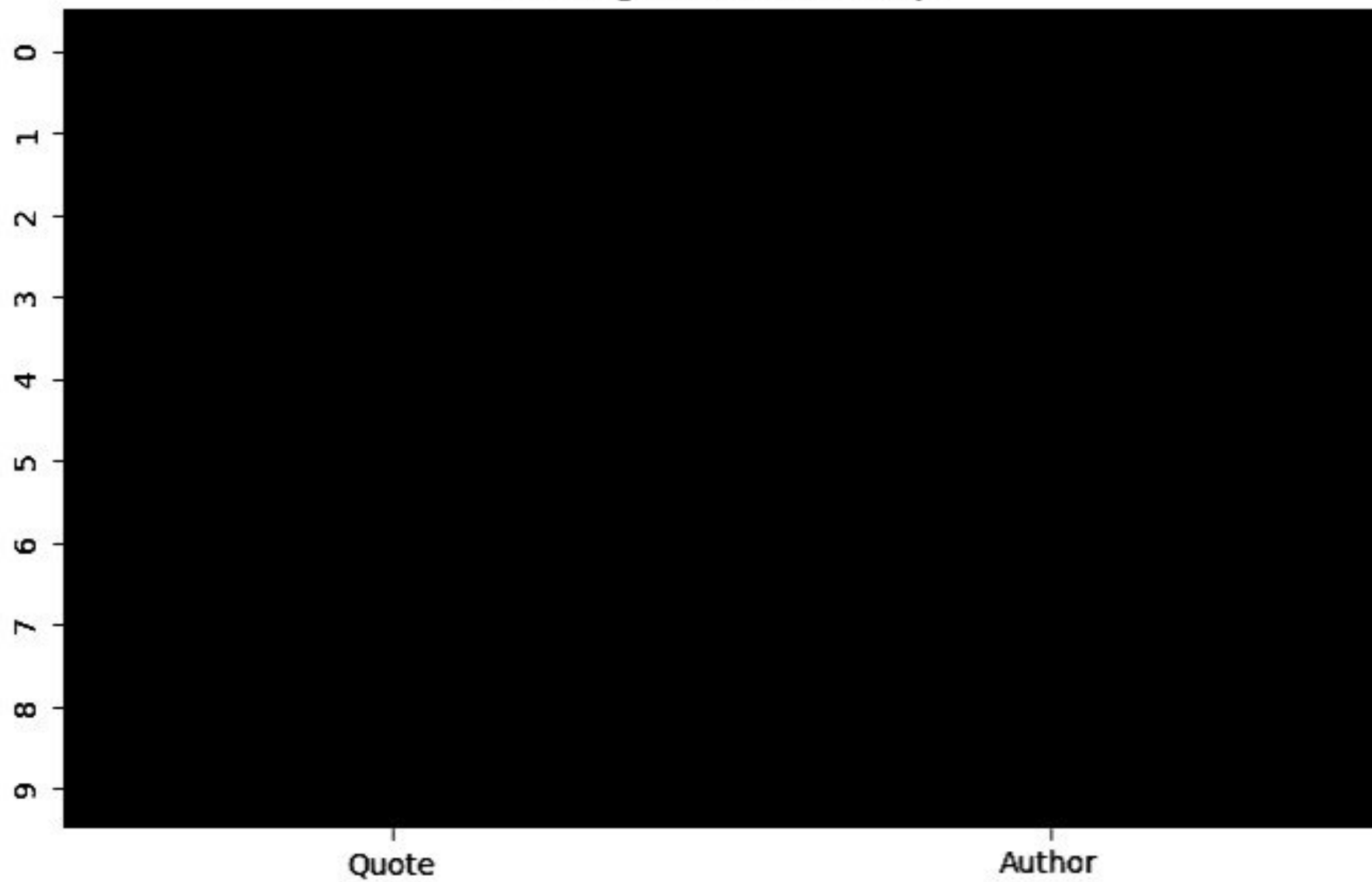
Missing Values:

Quote 0

Author 0

dtype: int64

Missing Values Heatmap



Statistical Summary:

	Quote	Author
count	10	10
unique	10	8
top	"The world as we have created it is a process ...	Albert Einstein
freq	1	3

Sample Questions:

1. Which category appears most frequently?
2. Are there missing values?
3. What patterns exist in the dataset?
4. Are there any unusual observations?

Duplicate Rows: 0

Data Quality Report

Missing Values:

Quote 0
Author 0
dtype: int64

Duplicate Records:

0

Data Types:

Quote object
Author object
dtype: object

EDA Completed Successfully!